

Homework for Chap 12

1. Calculate the plasmon energy for n-type GaAs with carrier concentrations of 10^{10} , 10^{15} and 10^{20} cm^{-3} .
2. Calculate the density of electrons at which $\hbar\omega_{\text{PL}} = \hbar\Omega_{\text{LO}}$ for ZnO and InAs, here $\hbar\omega_{\text{PL}}$ and $\hbar\Omega_{\text{LO}}$ are the plasmon energy and longitudinal optical phonon energy, respectively. Up to which temperatures are the electron gases degenerate?

$$1. \quad \omega_p = \sqrt{\frac{ne^2}{\epsilon_0 m^* \epsilon_{\infty}}}$$

$$m^* = 0.067 m_e$$

$$n = 10^{10} \text{ cm}^{-3} \quad \omega_p = 6.60 \times 10^9 \text{ rad/s} \quad E_f = \hbar\omega_p = 435 \text{ } \mu\text{eV}$$

$$n = 10^{15} \text{ cm}^{-3} \quad \omega_p = 2.09 \times 10^{12} \text{ rad/s} \quad E_f = \hbar\omega_p = 1.37 \text{ meV}$$

$$n = 10^{20} \text{ cm}^{-3} \quad \omega_p = 6.60 \times 10^{14} \text{ rad/s} \quad E_f = \hbar\omega_p = 4.35 \times 10^{-1} \text{ eV}$$

$$2. \quad \omega_p = \sqrt{\frac{ne^2}{\epsilon_0 \epsilon_{\infty} m^*}} = \Omega_{\text{LO}} \quad , \quad E_F = \frac{\hbar^2}{2m^*} (3\pi^2 n)^{\frac{2}{3}}$$

$$\textcircled{1} \text{ ZnO: } \epsilon_{\infty} = 3.7 \quad m^* = 0.24 m_e \quad \hbar\Omega_{\text{LO}} = 0.072 \text{ eV}$$

$$\Rightarrow n_{\text{ZnO}} = 3.34 \times 10^{18} \text{ m}^{-3} = 3.34 \times 10^{18} \text{ cm}^{-3}$$

$$\Rightarrow E_F = 0.034 \text{ eV}, T_F = \frac{E_F}{k_B} = \underline{\underline{394 \text{ K}}}$$

$$\textcircled{2} \text{ InAs } \epsilon_{\infty} = 12.3 \quad \hbar\Omega_{\text{LO}} = 0.029 \text{ eV} \quad m^* = 0.023 m_e$$

$$\Rightarrow n_{\text{InAs}} = 1.73 \times 10^{23} \text{ m}^{-3} = 1.73 \times 10^{17} \text{ cm}^{-3}$$

$$\Rightarrow E_F = 0.044 \text{ eV}, T_F = \frac{E_F}{k_B} = \underline{\underline{570 \text{ K}}}$$